



Comparing factor models across different market regimes: Evidence from China[☆]

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ABSTRACT

This study examines how cross-sectional return patterns in the Chinese stock market vary across market regimes by evaluating the performance of factor models. Employing the modified backward sup ADF method alongside the DF-GLS test, we identify distinct market phases over the past decade, including bubble, random walk, and extended downward periods. We compare eight prominent factor models within each regime and show that their relative performance varies systematically. During bubble periods, the behavior factor model by Danielet al. (2020) consistently outperforms others, suggesting that investor overconfidence, limited attention, and retail dominance play a central role in driving asset prices. In contrast, during efficient phases, the q -factor model by Houet al. (2015) exhibits superior explanatory power, highlighting profitability and investment as key fundamentals shaping cross-sectional returns in China. However, all models exhibit limited explanatory power during extended downturn periods, indicating substantial distortions in market pricing mechanisms.

1. Introduction

Over the past three decades, the Chinese stock market has become the world's second-largest, achieving a market capitalization of an impressive 95.4 trillion RMB by June 2022. Despite its considerable size, its relatively low correlation with global markets presents an attractive opportunity for international diversification (Hou et al., 2023). Distinct from more developed markets, it features a predominance of retail investors and an opaque information environment, which pose intriguing challenges for understanding the cross-sectional return patterns within this market (Liu et al., 2019; Hou et al., 2023).

Factor models play a pivotal role in understanding the cross-sectional return patterns and asset pricing mechanisms in stock markets (Liu et al., 2019; Hou et al., 2023). Since (Fama and French, 1993) introduce the seminal three-factor model, the asset pricing literature has expanded significantly. In particular, several theoretically motivated factor models have gained prominence, accommodating both traditional risk-based views and behavioral perspectives. Carhart (1997) extend (Fama and French, 1993)'s model to include a momentum factor. Hou et al. (2015), inspired by neoclassical q -theory in corporate finance, develop the q -factor model that incorporates investment and profitability factors alongside the traditional market and size factors. Fama and French augment their original three-factor model by including investment and profitability factors derived from the Dividend Discount Model (DDM) (Fama and French, 2015), and subsequently incorporate the momentum factor (Fama and French, 2018). From a behavioral finance perspective, Daniel et al. (2020) propose two factors (i.e., financing and post-earnings-announcement drift) to capture investors' overconfidence and limited attention which may lead to long- and short-horizon mispricing.

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While the robustness of the Fama–French five-factor model is well-established in developed markets (Fama and French, 2017), its effectiveness in emerging markets has shown mixed results. Additionally, though the significance of the momentum factor is well-documented in western markets, it does not necessarily apply to emerging markets like China (Asness et al., 2013). Importantly, the q -factor model has displayed promising performance. Through factor-spanning tests in the U.S. context, Hou et al. (2019) find that it dominates both the Fama–French five- and six-factor models, as well as the behavioral factor model by Daniel et al. (2020). Findings have been similar in the Thailand context (Charoenwong et al., 2021) and in the China context (Wang and Zhu, 2024; Li et al., 2024). However, the universality of the q -factor model's dominance has remained contested. Barillas and Shanken (2018) find that Fama–French's six-factor model is overwhelmingly superior to the q -factor and five-factor models. The findings in African stock markets also challenge the q -factor model's position of dominance among other models (Mbengue et al., 2023). Moreover, Hanauer et al. (2024) find that the four-factor model by Liu et al. (2019), specifically tailored for China, outperform both the Fama–French and q -factor models in Chinese stock market. In sum, these contrasting findings suggest that systematic risks driving excess returns vary in different equity market.

Notwithstanding fruitful results have been yielded in existing studies, the performance of prominent factor models across different market regimes remains ambiguous. Previous research often focus on establishing a universal asset pricing mechanism and implicitly assume stable return dynamics. This presumption sits uneasily with both theory and evidence (Cong et al., 2025). Theoretically, multi-factor models represent empirical approximations of the stochastic discount factor, which varies with business cycle states (Fama and French, 1993). If the pricing kernel is state dependent, then the set of relevant risk exposures and their associated premia should also vary across regimes. A time-invariant factor specification therefore risks misspecification when regime shifts are economically meaningful. Empirically, return predictability tends to cluster in specific market regimes rather than remain uniform over time (Henkel et al., 2011; Farmer et al., 2023), and the covariance structure and broader dynamics of returns display substantial instability (Moskowitz, 2003; Pukthuanthong et al., 2019). These findings motivate a regime-based evaluation of factor model performance. The Chinese stock market, with its recurrent bubbles and extended downturns (Chen et al., 2021), thus offer ideal laboratories for this kind of evaluation.

While a handful of studies have acknowledged the potential impacts of market regime switch, their explorations are predominantly confined to Fama–French factor models (Wang et al., 2021; Yu and Li, 2023). A critical limitation of these studies is that these models are developed on the Efficient Market Hypothesis (EMH), which assumes stock returns are unpredictable and stock prices follow random walks (Fama, 1970). Yet, this hypothesis fails to hold during bubble periods (Phillips et al., 2015). These theoretical and empirical challenges raise several fundamental questions regarding asset pricing in dynamic market environments: Is there truly a kind of universal asset pricing mechanism that effectively reflects diverse market conditions? Can any specific factor model consistently outperform others across various market scenarios? Will the recurrent bubbles and extended downturns in the Chinese market undermine the effectiveness of models based on the EMH? Can the four-factor model, tailored for the Chinese market by Liu et al. (2019), consistently demonstrate superiority across different market conditions as previously documented? If not, which model performs the best?

Against the backdrop, we aim to understand the cross-sectional return patterns in the Chinese stock market across different market regimes and shed light on the underlying asset pricing mechanisms through the lens of factor models. First, we employ the modified backward sup Augmented Dickey–Fuller method (mBSDF) by Phillips et al. (2015) and Wang et al. (2023) alongside the Augmented Dickey–Fuller test on Generalized Least Squares-detrended data (DF-GLS) by Elliott et al. (1996) to identify bubble and efficient periods. Subsequently, we compare the performance of prominent factor models across these different regimes using the wild-bootstrap GRS test as recommended by Giglio et al. (2021) to accommodate the potential issues caused by the time-varying volatility. The factor models under consideration include the three-, five- and six-factor models by Fama and French (1993, 2015) and Fama and French (2018) (FF3, FF5, FF6), the four-factor model by Carhart (1997) (FFC), the q -factor model by Hou et al. (2015) (HXZ4), the behavioral factor model by Daniel et al. (2020) (DHS3), and the three- and four-factor models (LSY3, LSY4) tailored to the Chinese stock market by Liu et al. (2019).¹ Finally, to validate our results, we apply the spanning test to analyze the relative performance of these models across different market regimes and conduct a battery of robustness checks.

Our contributions are threefold. First, we provide the first systematic identification and classification of stock market regimes in the China, documenting when the market exhibits bubble behavior, random walk dynamics, or extended downturns. This regime taxonomy offers a foundational framework for understanding China's distinctive market characteristics. Second, we contribute to the asset pricing literature by demonstrating the empirical limitations of universal factor models through regime-conditional evaluation. Our comparative analysis across the identified regimes reveals distinct underlying asset pricing mechanisms, challenging the widespread practice of full-sample evaluation and underscoring the importance of explicitly accounting for regime-dependent risk. Third, our work updates capital market research concerning the COVID-19 pandemic. Our findings reveal that while the behavioral factor model (DHS3) demonstrates superior performance during the pandemic's early phase, all candidate factor models exhibit limitations in capturing market dynamics during its later stages.

Our findings of this study can be summarized as follows. Firstly, we find that Chinese stock market has witnessed three bubble periods (i.e., 2014.11–2015.06; 2017.07–2018.02; 2020.06–2021.02), two intervals characterized by random walks (i.e., 2015.06–2017.07; 2018.02–2020.06), and two extended downturn periods (i.e., 2011.10–2014.11, 2021.02–2023.12) over the past decade.

¹ We differentiate our research from others by exclusively focusing on theory-motivated factor models to illuminate the underlying asset pricing mechanisms, thus excluding other prominent models such as those by Stambaugh and Yuan (2017), Barillas and Shanken (2018), and Kelly et al. (2019). The factor model by Stambaugh and Yuan (2017) is developed based on anomalies in the U.S. market; Barillas and Shanken (2018)'s approach is somewhat data-driven; and the latent factor model proposed by Kelly et al. (2019) still requires further investigation to enhance its interpretability.

These findings illustrate the alternate emergence of the bubble and random walk periods in the Chinese stock market. This possibly explains why the heterogeneity in sample period choices contribute to the ongoing lack of consensus regarding the best-performing factor model in this market (Jansen et al., 2021). Secondly, our results demonstrate that no factor model dominates universally, hence there is no universal asset pricing mechanism that consistently outperforms across different market conditions in China. Specifically, DHS3 model outperforms others during bubble periods, suggesting that speculative behavior primarily drives price dynamics during these times. In contrast, HXZ4 model demonstrates superior performance in efficient markets, which highlights the relevance of profitability and investment during these periods (Hou et al., 2023). This alludes to the critical role of a firm's growth potential and future investment opportunities in shaping asset pricing mechanisms. Finally, we discover that all tested prominent factor models are inadequate in explaining the cross-sectional return patterns in the extended downturn periods. The limitations of current models under conditions of persistent market decline suggest that such downturns significantly disrupt traditional pricing mechanisms. This underscores the need for further research into developing models that can better accommodate such adverse conditions. The spanning test and robustness check results validate our previous outcomes.

The remainder of this paper is organized as follows. Section 2 characterizes the distinctive features of Chinese stock market. Section 3 elaborates the factor construction and test portfolio formation, analyzing their summary statistics across diverse market regimes. Section 4 presents our main results. Section 5 reports the results of our robustness check. Section 6 concludes.

2. Alternating market regimes in the Chinese stock market

2.1. Characteristics of stock market bubbles

The Chinese A-share market has expanded dramatically, growing from only eight listings in December 1990 to more than 5000 today. Despite its impressive growth, the Chinese stock market exhibits several features that distinguish it from advanced markets, primarily due to its relatively short history (Leippold et al., 2022; Li et al., 2024). First, it has been well-documented that the A-shares market is dominated by retail investors (Jones et al., 2021). Most of them incline to exhibit speculative behaviors, which are highly possible to intensify market volatility. Second, the ongoing development of regulatory and trading institutions, including stringent short-selling restrictions and lengthy IPO approval processes, often creating inflationary or deflationary pressures, which might result in mispricing (Liu et al., 2019). These institutional and behavioral features, together with the market's sensitivity to rumors, imitation, speculation, and herding, make the Chinese stock market more vulnerable to bubbles and crashes than advanced markets (Yang and Ferrer, 2023).

To provide deeper insight, Fig. 1 plots the weekly closing prices of the China Securities Index 300 (CSI300) from January 2011 to December 2023, which serves as a proxy for overall market conditions.² We focus on this period for two reasons. First, the post-2007 sample provides a better playing field for factor model comparison in China, due to the completion of the split-share structure reform and the implementation of new accounting standards (Li et al., 2024). Second, we set 2011 as the starting year in order to exclude the 2007–2010 global financial crisis period. This also helps avoid a left-censoring problem in bubble detection. If a bubble episode had already emerged before 2007, initiating the test in 2007 would distort the identified origination and duration of the bubble episode. As shown in Fig. 1, pronounced volatility is evident, with significant surges and falls of prices around events such as the “2015 stock market crash” and the “COVID-19 epidemic”. These frequent volatile price swings vividly suggest recurrent explosive behaviors in the Chinese stock market. As noted by Moskowitz (2003), such features may affect the relevance of pricing factors. It is therefore important to identify bubble periods precisely and reassess the performance of established asset pricing models under those conditions.

We formally identify bubble episodes using the mBSDF procedure of Wang et al. (2023).³ The three detected bubble periods are shaded in gray in Fig. 1. The first bubble spans from November 16, 2014 to June 21, 2015. This episode coincides with the well-known “2015 Chinese stock market crash”. As widely discussed in the literature, it was primarily driven by policy-induced investor optimism and excessive market participation. During this period, the government actively promoted equity investment under the “China Dream” narrative, fostering unrealistic expectations of sustained market gains (Pan and Mishra, 2018). At the same time, a series of interest rate cuts increased market liquidity and encouraged risk-taking, particularly through margin trading (Yang and Ferrer, 2023). These conditions attracted a surge of inexperienced retail investors, whose speculative trading and herding behavior further amplified price distortions (Jones et al., 2021).

The second bubble runs from July 9, 2017 to February 4, 2018. Its emergence is closely related to contemporaneous market developments. In June 2017, MSCI announced that A-shares would be included in the MSCI Emerging Markets Index starting in June 2018. This milestone was widely interpreted as a signal of sustained foreign institutional participation, which significantly reinforced expectations of valuation re-rating and incremental capital inflows into the Chinese stock market (Li et al., 2024). The announcement was followed by rising trading activity and increasing foreign inflows (Zhang et al., 2021; Jiao et al., 2024), thereby creating favorable conditions for explosive price dynamics.

The third bubble spans from June 28, 2020 to February 21, 2021, during the early stage of the COVID-19 pandemic. This episode can be attributed to a combination of abundant liquidity, improving domestic fundamentals, and supportive policy signals. First, the

² The CSI300 tracks the performance of 300 leading A-share stocks listed on the Shanghai and Shenzhen Stock Exchanges and is widely regarded as a benchmark for the Chinese stock market.

³ Details of the method are provided in Appendix A.1.

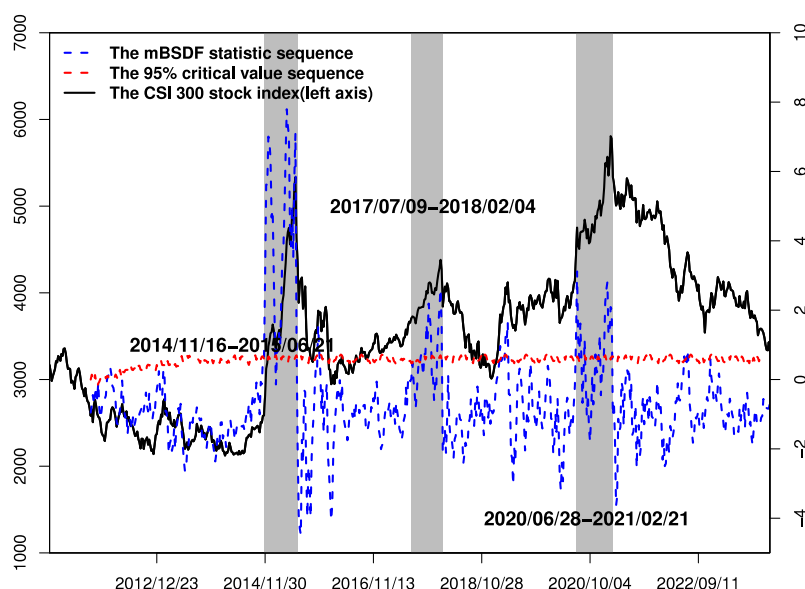


Fig. 1. The bubble periods in the CSI300 index.

Table 1

DF-GLS test results for non-bubble periods.

ADF statistics	2011/10/09–2014/11/09	2015/06/28–2017/07/02
	−2.850**	−1.908
	2018/02/11–2020/6/21	2021/02/28–2023/12/31
	−1.523	−2.786*

Note ***, ** and * indicate 1%, 5% and 10% significance level respectively, similarly hereinafter.

People's Bank of China implemented targeted liquidity injections to cushion the economic shock of the pandemic, reducing financing costs and increasing the amount of capital available for investment. Part of this liquidity likely flowed into the stock market and pushed prices upward. At the same time, China's rapid containment of the virus and subsequent economic recovery contrasted sharply with the prolonged difficulties faced by other major economies (Huang et al., 2023), strengthening investor confidence and attracting additional foreign participation in Chinese equities (Zhang et al., 2021). Continued policy support for capital market development further reinforced sentiment and encouraged speculative trading. Taken together, these conditions created favorable ground for another stock market bubble in China.

2.2. Distinguishing random walk and extended downturn regimes

Bubble episodes account for only a small fraction of the full sample. To obtain a more complete characterization of market regimes, we further classify the remaining non-bubble periods according to their underlying market dynamics. Specifically, after excluding the bubble episodes identified by the mBSDF procedure, we apply the DF-GLS test to the remaining subsamples to determine whether they follow a random walk and can therefore be classified as market efficient.⁴

The DF-GLS results reported in Table 1 show that, at the 5% significance level, the null hypothesis of a random walk cannot be rejected for the periods from June 28, 2015 to July 2, 2017 and from February 11, 2018 to June 21, 2020. These two periods are therefore classified as market efficient regimes. By contrast, for the periods from October 9, 2011⁵ to November 9, 2014 and from February 22, 2021 to December 31, 2023, the null of a random walk is rejected at the 5% and 10% levels, respectively.

These two non-efficient periods are more appropriately classified as extended downturn regimes rather than conventional trend-stationary episodes, as their dynamics are dominated by persistent downward trends. Based on Eq. (A.4), the estimated trend equations are $y_t = 2618 - 2.12t + y_t^*$ and $y_t = 5648 - 16.46t + y_t^*$, respectively. In both cases, the negative trend coefficients indicate sustained declines in prices. This feature is particularly pronounced in the latter period, during which the CSI300 fell from 5779 to 3329, broadly coinciding with the post-COVID-19 phase. To visualize this feature more clearly, Figure D.1 plots the fitted linear trend together with the CSI300 closing prices for the post-pandemic downturn period from February 28, 2021 to December 31,

⁴ Details of the DF-GLS procedure are provided in Appendix A.2.

⁵ The first period begins on October 9, 2011 rather than January 1, 2011, because we exclude the initial 39-week window required for bubble detection under the mBSDF procedure.

Table 2
Classification of market regimes.

Regime	Sample period
Extended downturn period I	2011/10/09–2014/11/09
Bubble Period I	2014/11/16–2015/06/21
Random Walk Period I	2015/06/28–2017/07/02
Bubble Period II	2017/07/09–2018/02/04
Random Walk Period II	2018/02/11–2020/06/21
Bubble Period III	2020/06/28–2021/02/21
Extended Downturn Period II	2021/02/28–2023/12/31

Note: Bubble periods are identified using the mBSDF procedure. The remaining periods are classified as random-walk or extended-downturn regimes based on the DF-GLS test and trend characteristics.

2023. The dashed trend line tracks the actual index path closely, suggesting that a linear downward trend captures a substantial part of market dynamics during this regime.

As a quantitative assessment, we adapt the logic of the coefficient of determination (i.e., R^2), which traditionally measures the variance in the dependent variable explained by the independent variables, as follows: $R^2 = 1 - SS_{res}/SS_{tot}$, where SS_{res} represents the residual sum of squares and SS_{tot} denotes the total sum of squares. For the purpose of this study, we slightly modify R^2 to specifically capture the influence of the downward trend by using $SS_{res} = \sum_t (y_t - \hat{\delta}_1 t)^2$ instead of the conventional definition. For the period from February 28, 2021 to December 31, 2023, the value of our modified R^2 is 72.59%, supporting our hypothesis that prevailing market conditions or macroeconomic factors have continuously exerted downward pressure on the index. The Ljung–Box test on the detrended series indicates significant ARCH effects, suggesting pronounced volatility clustering during this period.

Overall, the Chinese stock market alternates among bubble, market efficiency, and extended downturn regimes. The regime classification is summarized in Table 2. Motivated by these dynamics, we partition the full sample into distinct phases to facilitate a thorough evaluation of factor model performance across varying market regimes.

3. Data

3.1. Analysis of factors across different market regimes

Given these distinct market phases, we construct and analyze factors that could potentially capture the unique characteristics of each period. We obtain relevant data for all A-shares listed on the SSE and SZSE from the China Stock Market and Accounting Research database (CSMAR). Following Liu et al. (2019), we exclude “special treatment” (ST) stocks due to their trading restrictions and remove stocks (i) listed for less than six months or (ii) in abnormal status, trading days fewer than 120 in the last year or 15 in the last month. Moreover, when constructing factors proposed by Liu et al. (2019), we eliminate the bottom 30% of stocks ranked by market value following their practice.

We use weekly data instead of the traditional monthly data following Fama and French (1993) for the following considerations: on one hand, the monthly frequency data in our subsamples (i.e., divided into bubble and random walk periods) is too scarce, potentially leading to unreliable conclusions; on the other hand, daily data often exhibits excessive volatility, which can also undermine reliability. Moreover, we adopt the standard 2×3 sorting, following Fama and French (1993), and value-weighted procedure to construct all non-market factors, since Hou et al. (2023) and Li et al. (2024) document that equal-weighted scheme tends to disproportionately emphasize micro-caps with limited investment capacity. The compounded one-year fixed deposit rate is employed to proxy the weekly risk-free rate as standard. Since it is quoted on an annual basis, it is converted into a weekly rate as $r_{f,t}^{(w)} = (1 + r_{f,t}^{(a)})^{1/52} - 1$. Since Fama–French factors have been well-documented, we only briefly introduce the following factors here, with detailed construction procedures reported in Appendix C.

- *VMG*: value factor, measured by the ratio of a firm’s earnings to its market capitalization, see Liu et al. (2019).
- *r_{ROE}*: profitability factor, measured by return-on-equity (ROE), see Hou et al. (2015).
- *PMO*: sentiment factor, measured by abnormal turnover which is the ratio of the average daily turnover over the past 20 trading days to the average daily turnover over the past 250 trading days, see Liu et al. (2019). *PMO* reflects the high trading intensity of a stock, often driven by irrational optimism. Frequent trading activities tend to inflate stock prices beyond what fundamentals justify (Baker and Stein, 2004), thus capturing speculative motivations that are particularly pronounced in the Chinese market.
- *FIN*: financing factor, constructed by equally weighting the 1-year net-share-issuance (NSI) and 5-year composite-share-issuance (CSI), see Daniel et al. (2020). NSI is the log of the ratio of split-adjusted shares outstanding over a 1-year period (Pontiff and Woodgate, 2008), while CSI represents the 5-year growth rate in market equity subtract stock returns, in logs. The *FIN* factor aims to capture long-horizon mispricing driven by well-informed managers exploiting market timing opportunities through share issues or repurchases. Investors often overlook it due to their overconfidence, resulting in predictable return patterns.

- *PEAD*: post-earnings announcement drift factor, measured by a 4-day cumulative abnormal return surrounding the most recent quarterly earnings announcement date, see Daniel et al. (2020). The *PEAD* factor captures short-horizon mispricing arising from the investors' limited attention, which causes market's sluggish response to earnings surprises. DellaVigna and Pollet (2009) documents that this delayed response can lead to predictable future abnormal returns.

In light of the previous analysis, we further compare the summary statistics of the pricing factors reported in Table 3, together with the visual illustrations of selected factors shown in Fig. 2, across the market regimes identified earlier. Table 3 reports the descriptive statistics of weekly factor returns across various subsample periods, with statistical significance assessed using heteroskedasticity and autocorrelation robust (HAR) t test proposed by Sun (2013). Overall, significant factor premia are concentrated in a limited number of factors and periods, indicating strong regime dependence. This pattern is consistent with the limited time-series length in each regime and the well-documented difficulty of detecting anomaly premia in short samples (Jansen et al., 2021; Hou et al., 2023; Li et al., 2024). *MKT* is significant only in B1, with a mean reaches 2.72%. Value-related factors display limited but informative significance. The book-to-market-based *HML* factor is significant only in B3, whereas the earnings-to-price-based value factor *VMG* is significantly negative in R2 and in both extended downturn periods, reaching -0.61% in SP2. This suggests that earnings-based valuation signals may be more persistent under non-bubble market conditions. Profitability and investment factors are largely insignificant during bubble and random-walk periods, but become more visible in extended downturns, especially in SP2. By contrast, momentum shows weak overall performance across regimes.

Regarding the non-Fama–French factors, we derive three principal observations. First, the behavioral factors *PMO*, *FIN*, and *PEAD* exhibit distinctive patterns during bubble periods. Compared with the generally weak performance of Fama–French-style factors, behavioral factors show more frequent significance during bubble periods, consistent with the view that investor behavior and limited attention play an important role in speculative market states (Shu and Tan, 2022). Specifically, *FIN* is positive and significant in B1, while *PMO* and *PEAD* become significant in B2 and B3. The opposite signs of *PMO* and *PEAD* in these later bubble periods suggest that behavioral mispricing is not uniform across bubble episodes. The elevated volatility of these factors during bubble periods, particularly for *PMO* in Fig. 2, further supports their role in capturing speculative trading and behavioral pricing components.

Secondly, *PMO* emerges as the most robust factor across market regimes. It is insignificant only in B1, but becomes significantly negative in all remaining periods, with mean returns ranging from -0.48% in R2 to -0.86% in B3. This persistent performance, together with its pronounced fluctuations in Fig. 2, aligns with prior evidence that liquidity is a crucial driver of cross-sectional returns in the Chinese market (Liu et al., 2019; Hou et al., 2023; Li et al., 2024). The evidence therefore suggests that liquidity-related pricing effects are not confined to speculative bubbles, but appear across different market states in China's equity market.

In conclusion, the unique dynamics and complexities of China's equity market, as elaborated in Section 2, warrant a deeper examination of factor performance across diverse market conditions. The evidence indicates that *FIN*, *PEAD*, and especially *PMO* play important roles in capturing behavioral mispricing, particularly during periods characterized by speculative trading. Meanwhile, the results for the extended downturn periods show that some traditional valuation, profitability, and investment factors become more informative when market conditions deteriorate. Overall, factor performance in China's equity market appears highly state-dependent rather than governed by a single stable set of premia.

Third, *PEAD* exhibits more regime-dependent behavior than *PMO*. It is significant only in B2, B3, and R2, suggesting that the pricing effect captured by *PEAD* is more sensitive to market conditions. Its significance in B2 and B3 indicates that earnings-announcement-related drift is more evident in later bubble episodes, while its significance in R2 shows that this effect is not limited to bubble periods. The lack of significance in SP1 and SP2 suggests weaker evidence for *PEAD* during extended downturns, contrasting with the broader persistence of *PMO* across regimes.

3.2. Analysis of test assets across different market regimes

To gain insight into how asset pricing dynamics vary across market regimes, we investigate the performance of our test assets across these different contexts. The choice of test assets is critical for discriminating among competing factor models. The standard approach following Fama and French (1993) employs 5×5 characteristic-sorted portfolios. However, because these portfolios are constructed using characteristics closely linked to the candidate factors themselves, they embed a pronounced factor structure “by construction”, which may mechanically favor the corresponding model and reduce the discriminatory power of pricing tests (Jegadeesh et al., 2019; Harvey and Liu, 2021). At the same time, it has been well-documented that idiosyncratic noise in individual stocks will result in errors in variables (EIV) problems (Fama and MacBeth, 1973; Shanken, 1992). In response to these concerns, we use the 10 Level-1 sector indices of the CSI300 as our test assets, namely Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Health Care, Financials, Information Technology, Telecommunication Service, and Utilities. These sector portfolios offer sufficient aggregation to mitigate idiosyncratic noise while avoiding the mechanical factor structure embedded in characteristic-sorted portfolios, thereby providing a more balanced framework for evaluating competing factor models across different market regimes.

Table 4 presents the summary statistics of weekly excess returns of the 10 sector indices of the CSI300 across our subsamples. Three patterns emerge from the analysis. First, excess returns vary systematically across regimes. B1 displays broad-based positive excess returns across all sectors, ranging from 1.44% to 2.62% weekly. The later bubble periods, B2 and B3, are more mixed, although most sectors still record positive average excess returns. By contrast, the random-walk periods are closer to neutral. R1 is marginally negative on average, while R2 is more balanced between positive and negative sector returns. The two extended downturn periods also differ. SP1 remains mixed and close to zero, whereas SP2 displays a more broadly negative sector-level pattern.

Table 3
Descriptive statistics of the factor returns (%).

Statistics	MKT	SMB	HML	VMG	RMW	r_{ROE}	CMA	UMD	PMO	FIN	PEAD
Panel A: Bubble Period I (2014/11/16–2015/06/21) (B1)											
Mean	2.72	0.46	−0.19	−1.04	−0.32	−0.17	−0.21	−0.26	0.35	0.24	0.63
St. Dev.	4.61	3.64	2.51	3.39	1.51	1.65	1.24	2.82	2.81	1.11	1.34
<i>t</i> -Stats	4.43**	0.41	−0.38	−1.22	−1.21	−0.91	−0.65	−1.01	0.52	3.18**	1.14
Panel B: Bubble Period II (2017/07/09–2018/02/04) (B2)											
Mean	0.34	−0.81	0.25	0.56	0.10	0.32	0.11	0.01	−0.65	0.12	0.48
St. Dev.	1.48	1.63	1.48	1.67	1.19	0.85	0.93	1.63	1.70	0.72	1.00
<i>t</i> -Stats	2.19	−1.75	0.58	1.25	0.35	1.57	0.43	0.05	−3.64**	−0.57	3.25**
Panel C: Bubble Period III (2020/06/28–2021/02/21) (B3)											
Mean	1.01	−0.52	−0.30	0.35	0.35	0.32	0.34	0.23	−0.86	0.00	0.28
St. Dev.	2.86	1.77	1.32	1.66	1.66	0.85	1.63	2.19	1.62	0.58	0.84
<i>t</i> -Stats	−1.07	−1.75	−2.35*	1.49	1.77	2.10	1.21	1.55	−3.83**	−0.08	3.86**
Panel D: Random Walk Period I (2015/06/28–2017/07/02) (R1)											
Mean	0.05	−0.52	0.13	−0.06	−0.08	0.12	−0.04	−0.70	−0.64	−0.06	0.12
St. Dev.	4.05	1.77	3.13	3.36	0.83	1.46	0.94	2.28	2.95	1.11	1.22
<i>t</i> -Stats	0.36	−1.75	0.17	−0.16	−0.45	0.92	−0.31	−1.75	−7.20***	−1.81	1.44
Panel E: Random Walk Period II (2018/02/11–2020/06/21) (R2)											
Mean	0.21	0.07	0.13	−0.35	0.17	0.27	0.19	−0.18	−0.48	0.05	0.16
St. Dev.	2.86	1.45	1.13	1.78	1.14	1.17	1.00	1.52	1.65	0.59	0.70
<i>t</i> -Stats	0.73	−1.75	−2.13	−3.19**	0.74	1.52	0.83	−1.14	−2.51*	0.70	3.69**
Panel F: Extended Downturn Period I (2011/10/09–2014/11/09) (SP1)											
Mean	0.26	0.32	0.00	−0.35	−0.14	0.10	−0.05	0.03	−0.53	0.05	0.13
St. Dev.	3.47	3.65	0.27	3.56	0.12	0.79	0.31	4.90	1.51	0.40	1.26
<i>t</i> -Stats	1.78	2.15	0.10	−2.38*	−4.95**	1.42	−1.11	0.16	−5.42**	1.05	1.45
Panel G: Extended Downturn Period II (2021/02/28–2023/12/31) (SP2)											
Mean	0.14	0.34	0.00	−0.61	−0.17	−0.16	−0.25	−0.20	−0.78	0.00	0.00
St. Dev.	0.62	2.08	0.44	1.41	0.42	0.24	0.48	1.40	5.59	0.15	1.32
<i>t</i> -Stats	2.12	2.86*	0.01	−6.24***	−3.15*	−3.80**	−4.39**	−2.07	−3.98**	0.10	0.02

Note: This table reports the mean, standard deviation, and HAR *t*-statistic by Sun (2013) for weekly factors. Panels A–G report results for the B1, B2, B3, R1, R2, SP1, and SP2 periods, respectively. The definition of each factor is described in Appendix C. For brevity, we report summary statistics for the average SMB factor only, although multiple versions of SMB are constructed and employed in our analysis.

Second, return volatility differs substantially across regimes. B1 stands out as the most volatile bubble episode for several sectors, with standard deviations above five in Energy, Industrials, Financials, Telecommunication Service, and Utilities. Later bubble periods show less uniform volatility, but still display elevated volatility in selected sectors. This cross-regime variation indicates that the market-state classification captures not only differences in average returns, but also changes in standard deviation.

Third, the Ljung–Box statistics indicate more pronounced volatility clustering during bubble periods, especially in B1, where most sector indices reject the null of no ARCH-type dependence. This is consistent with the view that speculative episodes are often accompanied by stronger herding and momentum trading (Allen et al., 2024). The Jarque–Bera statistics also show departures from normality in several sectors and periods. Taken together, these results suggest that the standard i.i.d. normal error assumption underlying the Gibbons et al. (1989) framework cannot be taken for granted for these test assets, which is consistent with the concerns raised by Yu and Li (2023).

In summary, the analysis indicates that bubble periods are generally associated with stronger positive excess returns, higher volatility in selected sectors, and more pronounced volatility clustering. Random-walk and extended downturn periods display more muted and mixed return patterns, with SP2 showing the clearest negative sector-level tendency. These findings suggest that test-asset returns are strongly state-dependent and therefore motivate the evaluation of factor models across distinct market regimes rather than under a single full-sample pricing relation. They also suggest that benchmark models such as the Fama–French and *q*-factor models may face greater difficulty in capturing atypical return dynamics during volatile episodes. Finally, the evidence of non-normality and ARCH-type dependence in test-asset returns further supports the use of robust testing procedures rather than relying solely on the conventional GRS test under i.i.d. normal errors.

4. The performance of prominent factor models under different market regimes

In this section, we analyze the ability of eight different factor models to explain the average excess return of the 10 primary sector indices of CSI300 index across various market regimes (i.e., periods characterized by bubble, random walk, and extended downturn). The cross-sectional specifications of these models are:

$$r_{it} - r_{ft} = \alpha_i + \beta_i' f_t + u_{it}, \text{ for } i = 1, 2, \dots, N; t = 1, 2, \dots, T, \quad (1)$$

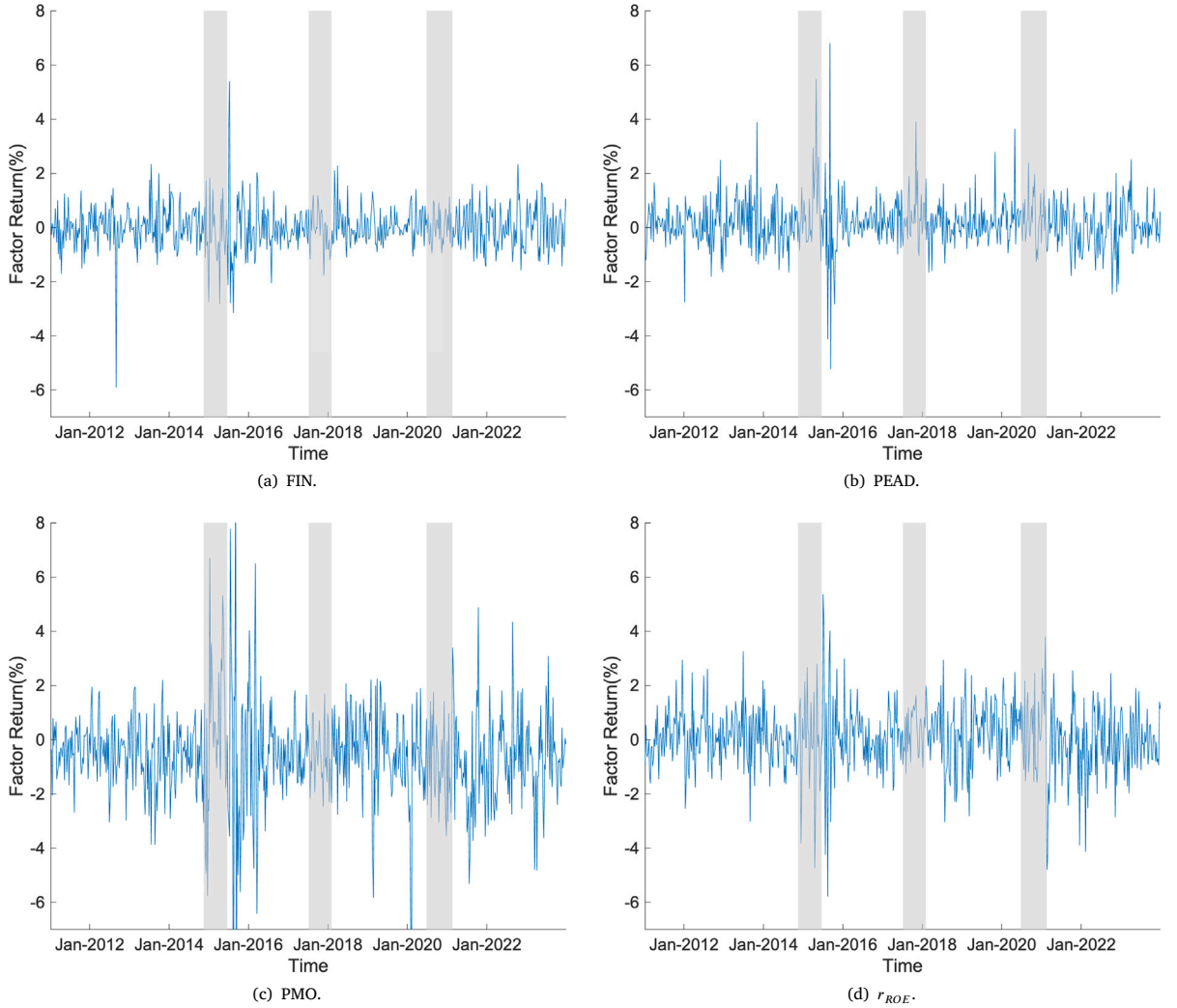


Fig. 2. Returns (%) of selected factors.

where r_{it} and r_{ft} denote the return of test asset i and the risk-free rate at week t , respectively. The vector $\mathbf{f}_t = (f_{1t}, f_{2t}, \dots, f_{Kt})'$ represents a $K \times 1$ vector of factors, which refers to FF3, FF5, FF6, FFC, HXZ4, LSY3, LSY4, and DHS3 models in this study. $\beta_i = (\beta_{i1}, \beta_{i2}, \dots, \beta_{iK})'$ is the corresponding factor loadings, and u_{it} is the error term.

Given the pronounced ARCH effects detected in test asset returns (see Table 4), we mainly rely on the wild-bootstrapped GRS test to evaluate the performance of factor models, for its reliability in finite samples and validity under weak assumptions regarding residuals (Gonçalves and Kilian, 2004; Kelly et al., 2019). Specifically, the GRS statistic and hypotheses are:

$$GRS = \frac{T - N - K}{N} [1 + \bar{f}' \hat{\Omega}^{-1} \bar{f}]^{-1} \hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha} \sim F_{N, T-N-K} \quad (2)$$

$$H_0 : \alpha_i = 0, \forall i = 1, \dots, N; \quad H_1 : \text{at least one } \alpha_i \neq 0.$$

T , N , and K are the number of time periods, test assets, and factors, respectively. As in Eq. (1), $\hat{\alpha}$ is the vector of the estimated intercepts, i.e., $\hat{\alpha} = [\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_N]'$, and $\hat{\Sigma}$ is the variance-covariance matrix of residuals ($\hat{\mathbf{u}}_t$). Moreover, $\bar{f}$ is the mean of the factors $\mathbf{f}_t$, and $\hat{\Omega}$ is the estimated covariance matrix.

The wild bootstrap procedure follows four steps: (1) estimate the parameters of the unrestricted model (1) and retain these estimated parameters for subsequent analysis; (2) generate the bootstrap residuals $\mathbf{u}_t^b := \hat{\mathbf{u}}_t \omega_t$, where $\{\omega_t\}_{t=1}^T$ are independent $N(0, 1)$

Table 4

Descriptive statistics of the excess returns (%) of the 10 CSI300 sector indices.

Statistics	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Bubble Period I (2014/11/16–2015/06/21) (B1)										
Mean	1.50	1.95	2.62	1.98	1.59	1.44	1.95	2.37	2.26	2.23
St. Dev.	5.35	4.62	6.23	4.11	4.21	3.46	6.35	4.78	6.36	5.03
Jarque–Bera	0.33	9.02***	1.54	12.52***	40.56***	37.04***	0.86	19.73***	18.20***	0.54
Q(10)	20.59***	42.68***	25.96***	94.91***	27.05***	41.80***	10.40	35.84***	46.06***	13.43
Panel B: Bubble Period II (2017/07/09–2018/02/04) (B2)										
Mean	0.68	0.69	0.07	0.47	1.35	0.22	0.64	0.23	−0.02	−0.17
St. Dev.	2.27	2.25	1.43	2.57	3.51	2.01	2.11	2.78	3.25	1.27
Jarque–Bera	2.60	1.28	0.16	1.15	1.50	0.84	0.05	0.09	1.29	0.75
Q(10)	17.35*	12.51	15.21	19.89**	18.42**	22.64***	16.42*	16.10	31.30***	14.86
Panel C: Bubble Period III (2020/06/28–2021/02/21) (B3)										
Mean	0.56	2.01	1.37	1.52	1.65	0.97	0.57	0.42	−0.27	0.24
St. Dev.	2.81	4.63	3.44	3.60	3.88	4.64	3.40	4.39	4.01	2.36
Jarque–Bera	1.44	1.01	0.68	1.29	1.58	2.10	0.86	0.38	0.01	2.63
Q(10)	19.40**	11.51	28.59***	14.43	13.31	43.57***	14.47	11.21	14.79	10.17
Panel D: Random Walk Period I (2015/06/22–2017/07/08) (R1)										
Mean	−0.34	−0.26	−0.42	−0.20	0.23	0.07	−0.10	−0.33	−0.15	−0.38
St. Dev.	3.61	3.94	4.27	3.59	3.06	2.95	3.28	4.63	3.87	3.36
Jarque–Bera	45.80***	35.33***	68.42	49.33***	14.46***	44.38***	25.66***	24.04	4.24	179.33***
Q(10)	9.87	16.96*	7.09	9.26	8.09	17.91*	6.67	12.22	5.00	12.69
Panel E: Random Walk Period II (2018/02/11–2020/06/21) (R2)										
Mean	−0.51	−0.20	−0.13	−0.03	0.42	0.34	−0.10	0.29	0.09	−0.11
St. Dev.	2.89	3.48	2.93	3.37	3.64	3.67	3.09	4.52	4.58	2.02
Jarque–Bera	75.25***	0.81	4.51*	8.47***	2.73	2.03	5.75*	0.25	14.73***	8.25**
Q(10)	4.54	3.51	6.04	4.56	10.62	10.62	5.36	3.21	10.79	6.55
Panel F: Extended Downturn Period I (2011/10/09–2014/11/09) (SP1)										
Mean	−0.41	−0.26	−0.07	0.10	−0.14	0.11	0.12	0.07	−0.10	0.10
St. Dev.	3.03	3.32	3.02	2.89	2.96	2.95	3.13	3.40	3.14	2.36
Jarque–Bera	3.24	1.79	1.09	0.23	3.53	0.63	8.48**	2.04	1.06	7.55**
Q(10)	5.10	6.03	3.76	4.69	17.15	13.10	3.33	6.77	8.34	8.71
Panel G: Extended Downturn Period II (2021/02/22–2022/12/31) (SP2)										
Mean	0.27	−0.36	−0.40	−0.43	−0.34	−0.48	−0.30	−0.31	−0.10	0.14
St. Dev.	3.38	3.27	2.80	2.95	3.72	3.43	2.58	3.30	3.02	2.64
Jarque–Bera	14.07***	0.72	2.54	11.71***	28.54***	0.09	3.75	2.96	6.45**	17.31***
Q(10)	13.17	9.80	17.92*	8.81	6.65	7.09	9.25	4.60	10.92	9.33

Note: This table reports weekly average excess returns from 10 CSI primary sector indices. Columns (1) to (10) represent Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Health Care, Financials, Information Technology, Telecommunication Service, and Utilities, respectively. Panels A–G report results for B1, B2, B3, R1, R2, SP1, and SP2, respectively. The Jarque–Bera statistic assesses normality of the return distribution. Q(10) denotes the Ljung–Box statistic with 10 lags for squared returns, which tests for ARCH-type dependence in the return series.

draws; (3) construct bootstrap samples $y_{it}^b = \hat{\beta}_i' f_t + u_{it}^b$ and compute bootstrap GRS statistics⁶; (4) calculate the bootstrap p -value (p^b) as the proportion of bootstrap statistics exceeding the actual GRS statistic.⁷

We primarily relied on the p^b -value, instead of the statistic, to assess the applicability of our candidate multi-factor models, since the factor number of candidate multi-factor models (i.e., K) differs. We also report three supplementary metrics following Fama and French (1993, 2015): $A|a|$, the average absolute value of intercepts, $A|t|$, the average absolute value of t statistics, and the mean adjusted R^2 (Mean adj. R^2). Generally, lower values of GRS statistics, $A|a|$, and $A|t|$ indicate higher applicability, while higher values of Mean adj. R^2 and p^b suggest more robust pricing ability. Additionally, we reported the results of the spanning test with HAR t statistics to support the robustness of our analysis.

4.1. The performance of prominent asset pricing models during bubble periods

Table 5 presents the performance of prominent factor models during bubble periods. Notably, the DHS3 model consistently demonstrate the highest p^b -values across all bubble periods, with values of 0.42, 0.72, and 0.14 during the B1, B2, and B3 periods, respectively. This indicates that the DHS3 model outperform others in pricing the cross-section returns during bubble periods. Moreover, the $A|a|$ of DHS3 rank the lowest in the B2 period, at 0.33, and the $A|t|$ ranked the lowest in the B1 and B2 periods, at 0.84 and 0.82, respectively, further affirming the superiority of the DHS3 model during the bubble periods.

⁶ Here, y_t is used to represent the left side (i.e., $r_{it} - r_{ft}$) of Eq. (1), and y_t^b is its bootstrap counterpart.

⁷ Our Monte Carlo simulations confirm the test's accuracy in maintaining appropriate rejection rates under both the null and alternative hypotheses, demonstrating enhanced performance, especially in power for closely adjacent alternatives, when employing the wild bootstrap. The results are available upon request.

Table 5
Performance of asset pricing models: bubble periods.

	FF3	FFC	HXZ4	FF5	FF6	LSY3	LSY4	DHS3
Panel A: Bubble Period I (2014/11/16–2015/06/21) (B1)								
GRS statistics	2.24	2.33	2.03	1.88	1.84	2.72	2.44	1.04
p^b value	0.02	0.02	0.09	0.06	0.12	0.02	0.02	0.42
$A a $	0.39	0.47	0.39	0.41	0.50	0.46	0.53	0.53
$A t $	0.89	1.03	0.90	1.01	1.22	0.84	1.74	0.84
Mean adj. R^2	0.77	0.77	0.80	0.83	0.83	0.73	0.75	0.69
Panel B: Bubble Period II (2017/07/09–2018/02/04) (B2)								
GRS statistics	1.43	1.69	2.54	3.15	3.54	1.58	2.04	0.68
p^b value	0.17	0.12	0.02	0.01	0.01	0.15	0.06	0.72
$A a $	0.39	0.38	0.42	0.42	0.37	0.37	0.44	0.33
$A t $	1.19	1.13	1.27	1.46	1.29	1.29	1.04	0.82
Mean adj. R^2	0.48	0.48	0.53	0.65	0.65	0.65	0.36	0.34
Panel C: Bubble Period III (2020/06/28–2021/02/21) (B3)								
GRS statistics	2.00	2.24	2.16	2.02	2.38	2.19	1.80	1.38
p^b	0.06	0.03	0.05	0.06	0.05	0.04	0.14	0.14
$A a $	0.43	0.43	0.46	0.38	0.42	0.43	0.32	0.50
$A t $	1.08	1.06	1.19	0.97	1.07	0.94	0.75	1.06
Mean adj. R^2	0.65	0.65	0.62	0.69	0.69	0.57	0.57	0.56

Note: This table reports the performance of eight factor models in explaining weekly returns of the 10 CSI300 sector indices during the three bubble periods. Reported metrics include the GRS statistic, wild-bootstrap p value p^b , average absolute intercept $A|a|$, average absolute intercept t -statistic $A|t|$, and mean adjusted R^2 . The best-performing model, defined by the highest p^b value, is highlighted in bold.

We attribute DHS3's superior performance during bubble periods to the interaction between its theoretical foundation and China's unique market microstructure. The DHS3 model explicitly incorporates two mispricing mechanisms, *PEAD* (reflecting limited attention) and *FIN* (capturing financing frictions), that become particularly salient during speculative episodes. Our factor analysis (Table 3) provides direct evidence: both *PEAD* and *FIN* exhibit elevated volatility and statistically significant premia specifically during bubble periods, with *PEAD* delivering mean returns of 0.48% in B2 and 0.28% in B3, while mean of *FIN* is 0.24 in B1. These empirical patterns align with well-established behavioral mechanisms. During bubble periods, two psychological biases intensify systematically. First, overconfidence escalates, manifested through surging trading volumes (Liao et al., 2022) and abnormal demand for speculative assets (Han et al., 2020). This overconfidence generates heterogeneous beliefs about asset values (Xiong and Yu, 2011), amplifying price deviations from fundamentals. Second, limited attention becomes more pronounced as investors fixate on price momentum and sensational news while neglecting fundamental information (Kumar, 2009). This selective information processing delays the incorporation of earnings surprises into prices, strengthening the *PEAD* effect.

China's retail-dominated market structure amplifies these mechanisms substantially. Retail investors, who constitute the majority of trading activity in Chinese markets, exhibit stronger preferences for lottery-like stocks and greater susceptibility to sentiment-driven trading compared to institutional investors (Kumar, 2009; Barber and Odean, 2008). Shu and Tan (2022) document that widespread financial illiteracy among retail participants makes *PEAD* particularly pronounced in China. Critically, retail participation intensifies during bubble periods (An et al., 2022), creating a self-reinforcing cycle: elevated retail activity → heightened overconfidence and limited attention → amplified mispricing → further retail engagement. This feedback loop magnifies mispricing to levels rarely observed in institutionally dominated markets.

Though LSY4 also incorporates a behavioral component through a turnover-based factor, Table 5 shows that DHS3 performs better on the key joint pricing metrics during bubble periods. A key reason is that turnover in LSY4 is a broad proxy for speculative activity, such as liquidity demand, disagreement, or general market activity. By contrast, DHS3 explicitly captures two mechanisms that are more tightly linked to bubble-period price formation.

The empirical dominance of DHS3 during bubble periods thus reflects a theoretical alignment: the model's mispricing factors (*PEAD* and *FIN*) capture precisely the behavioral distortions, overconfidence and limited attention, that are both theoretically expected and empirically verified to intensify during speculative episodes in retail-dominated markets. The pronounced volatility and consistent significance of these factors during bubble periods (Table 3) provide direct microstructural evidence supporting this interpretation.

4.2. The performance of prominent asset pricing models during efficient periods

Table 6 compares the performance of prominent factor models during random-walk periods, showing clear differences from their performance during bubble periods. Firstly, the Mean adj. R^2 values are significantly higher during random walk periods than in bubble periods, while $A|a|$ values are substantially larger during bubble episodes. These results suggest that traditional models struggle to capture return dynamics during bubble periods, likely because they are grounded in the EMH, which overlooks mispricing driven by irrational investor behavior. Additionally, HXZ4 performs best during the two random walk periods, as indicated by its highest p^b value (i.e., 0.15 and 0.03), as well as the low $A|t|$ (i.e., 1.30 and 1.24) and $A|a|$ values (i.e., 0.20 and 0.19). In contrast, the

Table 6
Performance of asset pricing models: periods of market efficiency.

	FF3	FFC	HXZ4	FF5	FF6	LSY3	LSY4	DHS3
Panel A: Random Walk Period I (2015/06/28–2017/07/02) (R1)								
GRS statistics	2.05	1.58	1.66	2.36	1.83	2.87	2.07	3.09
p^b value	0.06	0.10	0.15	0.04	0.07	0.01	0.05	0.00
$A a $	0.24	0.18	0.20	0.25	0.18	0.30	0.23	0.31
$A r $	1.48	1.08	1.30	1.62	1.20	1.86	1.08	1.80
Mean adj. R^2	0.79	0.81	0.81	0.82	0.83	0.78	0.79	0.75
Panel B: Random Walk Period II (2018/02/11–2020/06/21) (R2)								
GRS statistics	2.72	3.14	2.10	2.82	3.75	4.12	4.56	3.19
p^b value	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00
$A a $	0.21	0.22	0.19	0.21	0.22	0.25	0.22	0.27
$A r $	1.29	1.36	1.24	1.41	1.51	1.52	1.56	1.77
Mean adj. R^2	0.72	0.72	0.72	0.75	0.75	0.69	0.72	0.66

Note: This table reports the performance of eight factor models in explaining weekly returns of the 10 CSI300 sector indices during the periods of market efficiency. See table notes of Table 5 for the explanation of the metrics reported in this table.

DHS3 model, despite its strong performance during bubble periods, underperforms in these efficient market conditions, corroborating our earlier analysis regarding its context-specific strengths.

Our findings both corroborate and refine recent evidence on Chinese factor models. Li et al. (2024) identify LSY3, LSY4, and HXZ4 models as top performers across their entire sample period (2000–2019), with only HXZ4 maintaining dominance in the post-2007 subsample. Similarly, Hanauer et al. (2024) document HXZ4's superior performance though ranked behind the LSY3 and LSY4 models. However, our regime-specific analysis reveals that LSY3 and LSY4 consistently fail the wild bootstrap GRS test across most subsamples, an apparent contradiction explained by temporal market evolution. The perceived superiority of the LSY models has been largely established in studies incorporating pre-2007 data (Liu et al., 2019; Hanauer et al., 2024; Li et al., 2024), characterized by greater segmentation and inefficiency (Li et al., 2024). Following the 2007 structural break, marked by completion of the split, share reform and implementation of new accounting standards, the Chinese market's return-generating process fundamentally transformed, favoring models aligned with more efficient market mechanisms.

HXZ4's superiority in random walk periods reflects both theoretical coherence and institutional alignment with post-reform Chinese markets. The model's foundation in real investment theory provides the crucial theoretical advantage: firms invest until marginal revenue equals marginal cost, where marginal cost directly relates to market-to-book ratios. This creates strong theoretical correlation between HXZ4's *investment* factor and the *HML* factor, while empirical evidence shows *HML* becomes redundant in both U.S. Fama and French (2015) and post-2007 Chinese markets (Hsu et al., 2018). Notably, Hsu et al. (2018) attribute *HML*'s pre-2007 efficacy to non-tradable SOE shares being valued at book rather than market prices (Liao et al., 2014), a distortion eliminated by the split-share reform.

HXZ4's empirical advantage is further reinforced by institutional factors specific to China. The CSRC's endorsement of ROE as a critical profitability metric likely enhances the information content of HXZ4's *profitability* factor r_{ROE} . Moreover, the well-documented absence of *momentum* effects in the Chinese equity market (Cheung et al., 2015), possibly attributed to speculation-oriented retail investors prioritizing short-term gains (Li et al., 2024), further supports the applicability of the HXZ4 model over momentum-dependent models such as the FFC and FF6 models in this market. Finally, HXZ4's focus on *profitability* and *investment* factors aligns with Chinese retail investors' documented preference for growth stocks, potentially leading to overvaluation of glamor stocks (Hsu et al., 2018) that these factors effectively capture.

The empirical dominance of HXZ4 during random-walk periods thus reflects a similar theoretical alignment. In relatively efficient market states, pricing is more likely to be organized around firms' fundamentals rather than mispricing-related channels. HXZ4 performs well in such environments because its profitability and investment factors are more directly tied to these fundamentals. This advantage is particularly relevant in the post-2007 Chinese market, where institutional reforms reduced earlier distortions and strengthened the informational role of profitability- and investment-based signals.

4.3. The performance of prominent asset pricing models during extended downturn periods

Table 7 reports the results for the extended downturn periods. The GRS statistics are large in both SP1 and SP2, but the pattern is particularly striking in SP2, which corresponds to the post-COVID-19 period. The SP2 results suggest that established factor models exhibit substantially weaker pricing performance during this unusual market state. As discussed in Section 2.1, SP2 is characterized by pronounced market inefficiency and a dominant downward trend, with a GLS coefficient of -16.46 that accounts for 72.59% of the index variation, as shown in Figure D.1. Strict restrictions during this period, such as lockdowns, suppressed domestic consumption and disrupted normal business operations, logistics, and distribution networks (Goldstein et al., 2021). Under such conditions, the usual risk-return relation may have been overwhelmed by macroeconomic shocks and persistent downward pressure, making traditional factor decompositions less effective. This evidence suggests that, during extreme structural shocks, market-wide macroeconomic forces may dominate cross-sectional pricing patterns, thereby reducing the discriminatory power of multi-factor models (see Table 7).

Table 7
Performance of asset pricing models: periods of extended downturn.

	FF3	FFC	HXZ4	FF5	FF6	LSY3	LSY4	DHS3
Panel A: Extended Downturn Period I (2011/10/09–2014/11/09) (SP1)								
GRS statistics	5.32	5.71	4.99	5.59	6.38	6.02	3.43	6.85
p^b value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
$A a $	0.26	0.26	0.21	0.21	0.21	0.30	0.25	0.33
$A r $	2.17	2.22	2.43	1.82	1.89	2.41	1.85	2.43
Mean adj. R^2	0.72	0.73	0.73	0.75	0.76	0.70	0.71	0.73
Panel B: Extended Downturn Period II (2021/02/22–2023/12/31) (SP2)								
GRS statistics	7.04	7.53	6.04	6.40	6.88	12.06	9.25	6.04
p^b value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
$A a $	0.27	0.27	0.25	0.27	0.27	0.44	0.39	0.42
$A r $	1.63	1.65	1.48	1.66	1.69	2.29	1.88	2.34
Mean adj. R^2	0.56	0.57	0.54	0.58	0.58	0.49	0.48	0.44

Note: This table reports the performance of eight factor models in explaining weekly returns of the 10 CSI300 sector indices during the periods of extended downturns. See table notes of Table 5 for the explanation of the metrics reported in this table.

5. Robustness checks

5.1. Robustness of market regime classification

Accurate regime classification is important for the reliability of our conclusions. We assess the robustness of the regime classification along two dimensions. First, bubble identification is re-examined under several alternative mBSDF specifications, including the 90% and 99% critical-value specifications, the alternative modified-parameter specification ($K = \lfloor T^{1/5} \rfloor = 4$), and the alternative minimum-window specification ($f_0 = 0.01 + 1.8/\sqrt{T}$). Second, the KPSS test proposed by Kwiatkowski et al. (1992) is used as a complementary check for the classification of the non-bubble periods.

The detailed results are reported in the Online Appendix D.1. Table D.1 shows that all alternative mBSDF specifications identify the same three bubble episodes. Relative to the baseline specification, the estimated bubble durations differ by at most one week, except under the 99% critical-value specification, which yields somewhat shorter 2017–2018 and 2020–2021 bubbles. Table D.2 reports the KPSS results based on the baseline period partition. At the 5% level, stationarity is rejected for the two random-walk periods, supporting their classification as weak-form efficient regimes, while it cannot be rejected for the other two periods. Overall, the results confirm the robustness of the regime classification.

5.2. Performance comparison under different market regimes with buffer periods

To examine whether the regime-based results are driven by potentially ambiguous observations around state boundaries, we exclude observations within one week after and before each identified regime boundary and re-estimate model performance. The detailed results are reported in the Online Appendix D.2, show that the main regime-dependent patterns remain broadly intact.

For bubble periods, model with behavioral components remain the most competitive after excluding the boundary observations. DHS3 continues to deliver the highest bootstrap p^b -values (0.32 in B1 and 0.38 in B2). In B3, though, DHS3 shows the lowest GRS statistics at 1.41, LSY4 attains the highest p^b -value, at 0.25, and also yields the lowest $A|a|$ and $A|r|$. For random walk periods, the results continue to favor HXZ4 model. In R1, HXZ4 attains a joint-highest p^b -value of 0.16, and in R2 it records the highest p^b -value of 0.02. For extended downturn periods, the conclusions are unchanged. All models are strongly rejected in both SP1 and SP2, with p^b equal to zero throughout. Overall, the buffer-period results strengthen the credibility of the regime-based analysis. Excluding the transition weeks does not overturn the main findings.

5.3. Factor-spanning tests

To further validate the regime-based model comparison, we conduct factor-spanning tests using HAR-based inference by Sun (2013) to account for heteroskedasticity and serial dependence. The detailed results are reported in the Online Appendix D.3. The spanning tests are designed to examine whether the factors of the best-performing models contain incremental pricing information beyond the alternative benchmark models. Specifically, we assess whether the DHS3 factors can be spanned by other models during bubble periods and whether the HXZ4 factors can be spanned by other models during random-walk periods.

The results are broadly consistent with the main wild-bootstrap GRS evidence. During bubble periods, DHS3 factors provide incremental pricing information, especially through the behavioral channels captured by *PEAD* and, to a lesser extent, *FIN*. This supports the interpretation that limited attention and overconfidence become more relevant in speculative market states. During random-walk periods, the HXZ4 factors retain additional pricing information relative to several benchmark models, particularly through the size and profitability-related components. Overall, the factor-spanning evidence confirms that no single factor model dominates across all market regimes. Instead, asset pricing performance in the Chinese stock market is regime dependent.

6. Concluding remarks

This study provides the first systematic regime-based evaluation of prominent asset pricing models in the Chinese stock market. Using the mBSDF method, we identify three bubble periods (2014.11-2015.06, 2017.07-2018.02, and 2020.06-2021.02), two random walk periods (2015.06-2017.07 and 2018.02-2020.06), and two extended downturn period (2011.10-2014.11, 2021.02-2023.12) characterized by persistent downward trends. Given the pronounced ARCH effects and non-normality detected across these regimes, we employ wild bootstrap GRS tests to compare eight prominent asset pricing models. Our findings reveal striking regime-dependent patterns: the DHS3 model demonstrates superior performance during bubble periods, the HXZ4 model dominates during efficient market phases, while all models exhibit a universal decline in pricing ability during the extended downturn. Spanning tests corroborate these regime-specific patterns, confirming that superior models in each regime possess unique pricing information unexplained by alternative frameworks.

Our research diverges from the typical focus of empirical asset pricing literature, which often overlooks specific market characteristics and seeks a universal pricing mechanism applicable across all markets and periods. Our investigation yields three vital insights. First, the DHS3 model consistently outperforms alternative specifications across all bubble periods, suggesting that mispricing during speculative episodes is primarily driven by behavioral biases—specifically, overconfidence and limited attention. The model's *PEAD* and *FIN* factors exhibit elevated volatility and statistically significant premia during these periods (Table 3), with spanning tests revealing substantial alphas unexplained by traditional factors. This superior performance reflects a confluence of factors unique to China's market structure: retail investor dominance amplifies behavioral biases, while explosive price dynamics create self-reinforcing feedback loops that magnify mispricing beyond levels observed in institutionally dominated markets. The DHS3 framework effectively captures these psychological mechanisms, explaining its theoretical consistency with bubble-period dynamics.

Second, the HXZ4 model emerges as the dominant specification during random walk periods, underscoring the primacy of *investment* and *profitability* factors in explaining cross-sectional returns when markets operate efficiently. This finding corroborates (Li et al., 2024)'s evidence of HXZ4's superiority over LSY models in the post-2007 period and provides empirical support for Hou et al. (2019)'s theoretical argument regarding *value* factor redundancy. The model's dominance reflects both theoretical coherence — investment theory directly links marginal costs to market-to-book ratios, suggesting correlation between *investment* and HML factors — and institutional alignment with post-reform Chinese markets. The 2007 structural reforms, particularly the completion of split-share reform and implementation of new accounting standards, fundamentally altered the return-generating process, diminishing traditional HML effectiveness while elevating the importance of forward-looking metrics. HXZ4's emphasis on *profitability* (r_{ROE} , explicitly endorsed by the CSRC) and *investment* aligns with Chinese retail investors' documented preference for growth stocks, while the well-established absence of *momentum* effects in China naturally favors HXZ4 over momentum-dependent alternatives like FFC and FF6.

Thirdly, all factor models lost their pricing powers during the extended downturn periods. This is evidenced by exceptionally large GRS statistics and significantly low Mean adj. R^2 values. We propose that the persistent downward trend due to the economic downturn dominated the market dynamics, resulting in substantial distortions in market pricing mechanisms.

These findings carry significant implications for both asset pricing theory and regulatory practice. Theoretically, our results demonstrate that systematic risk varies fundamentally across market regimes in emerging markets, rendering universal pricing frameworks inadequate. The stark regime-dependent performance, DHS3 excelling during speculative episodes, HXZ4 dominating efficient periods, and all models failing during severe downturns, underscores that effective asset pricing requires explicit recognition of state-dependent risk structures. This challenges the prevailing research paradigm that seeks single models applicable across all conditions and periods. For practitioners and regulators, our evidence highlights two critical considerations: behavioral biases amplify dramatically during speculative episodes in retail-dominated markets, necessitating enhanced monitoring of sentiment-driven dynamics; and traditional pricing mechanisms exhibit substantial vulnerability to structural shocks, suggesting the need for alternative risk assessment frameworks during crisis periods.

CRedit authorship contribution statement

Xinyu Wang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shaoping Wang:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Hao Feng:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.pacfin.2026.103234>.

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